

dSIBP: a package for generating IBP seeds for QFT in de Sitter spacetime

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ABSTRACT: dSIBP is a MATHEMATICA package. It generates canonical integration-by-parts (IBP) seed relations for the Feynman integrals of quantum field theory in de Sitter spacetime. The input is a Feynman graph with massive and massless propagators of the four Schwinger–Keldysh types. The package builds the corresponding integral family with structured integer labels. It applies the equation-of-motion and endpoint-state normalizations of the loop-integral formalism of de Sitter field theory. It then emits canonical seed equations in which every discrete label has been reduced to a finite basis. The seeds are exported through a backend-neutral linear-system format to the external reducers Kira and Rational Tracer. The reduced master integrals can be differentiated to obtain differential equations in a single command. The package covers arbitrary loop order and arbitrary topologies. It supports the parity-split IBP system specific to de Sitter time integrals, and it provides a parallel tree-level track that produces differential equations without any loop momentum. We describe the notation, the public interface, and worked examples ranging from the sunrise graph to a mixed loop-plus-tree family with eighty-one master integrals.

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1 Introduction

Correlation functions of quantum fields in de Sitter spacetime are computed in the in-in (Schwinger–Keldysh) formalism. In this formalism the time integrals run over two branches, and internal lines come in massive and massless variants that connect equal or opposite branches [1]. In momentum space every graph reduces to scalar integrals over the conformal times $\tau_v < 0$ of its vertices. The integrand is a product of plane waves $e^{\mp i E_v \tau_v}$ and of Hankel functions. These time integrals differ from the loop integrals of flat-space field theory. Under IBP reduction [2–5] they produce linear relations among integrals whose discrete labels count both propagator powers and the powers of the Hankel functions at the endpoints of each propagator. The dimension of this label space grows rapidly with the loop order. The reduction therefore becomes the dominant computational cost of any high-loop calculation in de Sitter space.

The systematic IBP structure of de Sitter time integrals was developed in refs. [6–8]. Single-vertex and two-vertex families admit closed-form IBP relations. Multivariate hypergeometric techniques solve the two-vertex massive families [7]. The loop-momentum

formalism with a parity-split IBP system covers general graphs [8]. In this formalism the IBP identities take a canonical form in which every discrete label is confined to a small finite set. The full linear system of a graph is therefore fixed once the topology, the propagator types and a few kinematic declarations are given. Writing these canonical seeds by hand is laborious and error-prone already at one loop. Each propagator carries three structured labels. Massive lines obey second-order equations of motion. Massless same-branch propagators must be treated as a quotient over endpoint states.

dSIBP is a MATHEMATICA package that automates this step. The user declares a Feynman graph by its vertices, by the four Schwinger–Keldysh types of its internal lines, and by its external momenta. The package then performs the following tasks.

- It validates the kinematic declarations, namely the loop momenta, the no-loop external magnitudes and their dynamical rules. It reports whether the user lists are complete or over-complete.
- It constructs the integral family J with structured labels for propagator powers, endpoint states and user-defined irreducible scalar products (ISPs).
- It generates canonical IBP seed equations from the time generators at all active vertices and from the $L(L+K)$ loop-momentum generators. The massive endpoint states are restricted to the proven representative set $\{0, 1\}$.
- It sweeps the continuous labels over a user-specified envelope and assembles a sealed, backend-neutral linear system.
- It exports the system to Kira [9, 10] or Rational Tracer [11] for the actual reduction, and it imports the results back.
- It differentiates the reduced master integrals with respect to the independent kinematic variables. This yields differential equations, optionally in $d \log$ form. Euler scaling identities provide an independent check of the result.

The package generates and serializes IBP relations, but it never runs a reduction itself. The reduction is delegated to external programs. This separation keeps the package independent of any particular reduction engine. A complementary tree-level track provides seed relations for single- and multi-vertex time-only families. These seeds can be reduced internally to differential equations and $d \log$ forms without any loop momentum. The track also interfaces with the package MadStree [12], which evaluates tree-level cosmological correlators numerically. Related structures for cosmological loop integrands appear in the kinematic-flow program [13–15].

All formulas used by the package are derived in refs. [6–8]; the package adds no new derivation. The purpose of this paper is therefore to state the notation in a complete and self-contained form (section 2), and to describe the public interface (sections 3 and 4). Every symbol that appears in the output of the package is traced back to its definition. The interface described below is that of the current public release.

Statement on AI use. The notation, the functionality and the interfaces of `dSIBP` were designed by the author. The author supervised a number of different agent programs and large language models that implemented the code and the documentation. The author also designed the validation tasks that establish the reliability of the package. All of these tasks are cross-validations of computed results against independent data. They cover the IBP relations, the differential equations and the scaling identities as follows.

- *Massive two-vertex family.* The general IBP seeds and the differential equations are checked against the analytic results of ref. [7]. The family has five normalized masters. The three 5×5 differential-equation matrices agree entry by entry, 25/25 exact in each matrix. The vertex basis and the ordering of the masters follow ref. [6]. This task validates the IBP relations and the differential equations.
- *Massive three-vertex tree family.* The formulas were first derived independently by an agent, which generalized an earlier hand-written two-vertex implementation. They were checked against ref. [8] and only then compared with the output of the package. This task validates the tree-level IBP relations and the tree-level differential equations.
- *One-loop bubble, full pipeline.* The IBP system is reduced by an external Kira run, which closes with no unreduced target and returns nineteen masters. The two 19×19 differential-equation matrices keep both derivative variables symbolic. At the recorded exact point they reproduce the matrices of ref. [8] entry by entry, 361/361 for each of the three variable conventions. The scaling identities hold on the reduced basis: 33/33 source-level identities and 19/19 active-basis checks pass symbolically. This task validates the IBP relations, the differential equations and the scaling identities.
- *Mixed bubble-plus-tree family, full pipeline.* The family has eighty-one master integrals. The external Kira reduction closes with no unreduced target. The six 81×81 differential-equation matrices are free of residual symbols. All 81/81 source-level Euler scaling and IBP identities vanish exactly. These identities are independent of the numerical reduction. The reduction itself carries a scaling certificate at a recorded exact point. This task validates the IBP relations, the differential equations and the scaling identities.
- *General formulas against naive sampling.* The package substitutes the derived general formulas instead of solving IBP systems again. Its output was compared with reductions of naively sampled raw IBP seeds of the same families. The IBP relations and the differential equations agree. This task validates the IBP relations and the differential equations.
- *Tree-level differential equations against MadStree.* The direct $d \log$ route and the naive IBP route of the package give the same result. In addition, for a mixed three-vertex tree family with fifteen masters, the parameter derivatives of `dSIBP` were transported into MadStree [12] for the five variables $k_1, k_2, k_3, sE1, sE2$. All seventy-five derivatives closed, and the five 15×15 differential-equation matrices agree entry by entry with the

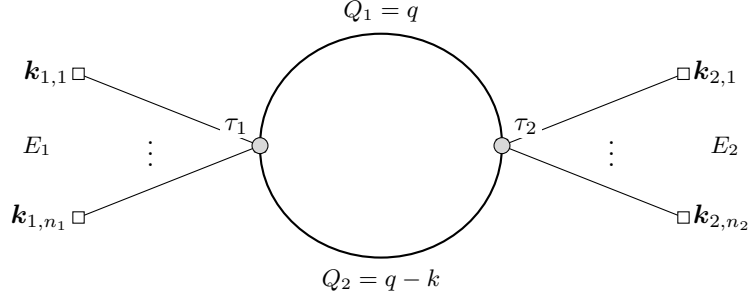


Figure 1. Feynman rules of a de Sitter bubble with general branch signs, drawn as gray vertices. The vertex energies E_1, E_2 and the conformal times τ_1, τ_2 are written at the vertices. The external legs end in open squares and carry the momenta $k_{1,1}, \dots, k_{1,n_1}$ and $k_{2,1}, \dots, k_{2,n_2}$. The two internal lines carry the loop momenta q and $q - k$.

direct $d \log$ formula route of MadStree. This task validates the tree-level differential equations.

The package is available at <https://github.com/J7C/dSIBP-Packages>.

2 Notation and conventions

2.1 Schwinger–Keldysh Feynman rules

A de Sitter Feynman graph in the in-in formalism consists of vertices v with conformal time $\tau_v < 0$ and Schwinger–Keldysh branch signs $\sigma_v = \pm$. Each vertex with external energy E_v carries the phase factor

$$e^{-i\sigma_v E_v \tau_v}, \quad (2.1)$$

and the time integrations run over the full interval $\tau_v \in (-\infty, 0)$. An internal line e that connects the vertices u and v is one of four types. It is massive or massless, and it is full (same branch, $\sigma_u = \sigma_v$) or cross (opposite branch, $\sigma_u = -\sigma_v$). The four propagator types have distinct Heaviside structures and different kernels. The massive full and cross propagators differ by a factor i , and the two massless cross propagators likewise differ by a factor i . The massless full propagator is the sum of its two ordered halves. In our figures a circle denotes a vertex, and its fill colour encodes the branch sign: black is $\sigma_v = +$, white is $\sigma_v = -$, and gray leaves the sign general. The conformal time τ_v is written below the vertex and the vertex energy beside it. External legs end in small open squares. The energy E_v of a vertex is the sum of the magnitudes of all external momenta routed into v , plus any vertex energy parameter z_v . Figure 1 shows these rules for the bubble input of the package. Its two vertices are drawn in gray because their branch signs are general. Vertex 1 carries the external momenta $k_{1,1}, \dots, k_{1,n_1}$ and vertex 2 carries $k_{2,1}, \dots, k_{2,n_2}$. The internal momenta are q and $q - k$.

The package distinguishes the four internal-line types as $\langle \text{massive}, \text{massless} \rangle \times \langle \text{full}, \text{cross} \rangle$. Figure 2 shows the resulting graphical conventions. Every cyclic line of the graph

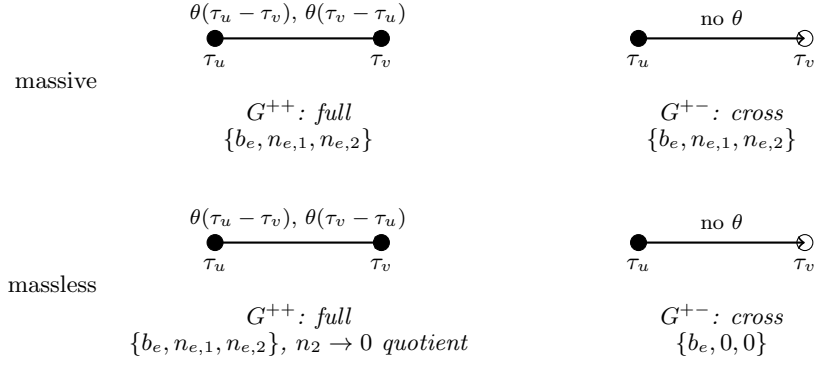


Figure 2. The four internal-line types and their label packs. The fill colour of a vertex encodes its branch sign: black is $+$ and white is $-$. The conjugate assignments $--$ and $-+$ give the same two line types. Massive and massless full lines carry $\{b_e, n_{e,1}, n_{e,2}\}$, with the massive endpoint states restricted to $\{0, 1\}$. Cross lines carry $\{b_e, 0, 0\}$. Massless full lines are the quotient of the two ordered endpoint states, with $F_{01} = -F_{10}$ and $F_{11} = -F_{00}$.

becomes a propagator of the integral family. Non-cyclic lines attached to the loop skeleton are *fixed lines*. They carry only a magnitude power and no loop-momentum denominator. Massive lines whose endpoints become coincident under a contact operation may be *shrunk*. Shrunk lines survive only through a structured prefactor of the sector.

2.2 The integral family

For a sector s , that is, a subset of propagators retained as denominators, a generic member of the integral family is a time-ordered integral over the vertex times,

$$J[s, \{a_v\}, \{B_e, n_{e,1}, n_{e,2}\}, \{\rho_r\}] = \int_{-\infty}^0 \prod_v d\tau_v \prod_v (-\tau_v)^{a_v + a_{0v}} e^{-i\sigma_v E_v \tau_v} \times \prod_{e \in s} \mathcal{G}_e(\tau_u, \tau_v)^{B_e} \prod_e H_e^{n_{e,1}} \bar{H}_e^{n_{e,2}} \prod_r \rho_r^{i_r}. \quad (2.2)$$

Here $e = (u, v)$ runs over the lines of the sector, $\mathcal{G}_e$ is the propagator of the line e , and $H_e, \bar{H}_e$ are the Hankel-function blocks attached to the two endpoints of a massive line. In the package the family is written in the three-slot schema

`J[sectorKey_String, linePacks_List, ispList_List]`

The first slot identifies the sector by the retained propagators. The second slot holds one pack per line, in the root line order.

- A cyclic full line contributes $\{b_e, n_{e,1}, n_{e,2}\}$: the propagator power b_e and the Hankel powers at the two endpoints.
- A cyclic massless cross line contributes $\{b_e, 0, 0\}$.
- A fixed line contributes $\{\text{"F"}, n_{e,1}, n_{e,2}\}$: no denominator power, only the magnitude powers of the line.

- A shrunk line contributes $\{b_{S,e}\}$, the magnitude power kept by the sector prefactor.

The third slot lists the ISPs with their integer indices i_r .

The zero points of the labels are fixed by the package and cannot be changed by the user. For the h -preset dynamics the vertex time powers have $a0_v = 2\nu_{\text{ref}}$, and the massive propagator powers have $b0_e = -2\nu_e$. The *actual* powers that enter the integrand are therefore $A_v = a_v + a0_v$ and $B_e = b_e + b0_e$, while the package-level labels $a_v, b_e, n_{e,i}$ run over small non-negative integers. These conventions follow refs. [7, 8]. They make the IBP shifts uniform: the generators shift the package-level labels by ± 1 and never touch the zero points.

Massive endpoint states and equations of motion. A massive line e with mass parameter ν_e and the h -preset $P_h(x) = (2\nu_e + 1)/x$, $Q_h(x) = 1$ has Hankel blocks at each endpoint that span a two-dimensional space. The space is indexed by the state $n_{e,i}$. The second-order equation of motion gives, at $n_{e,i} = 2$, the exact relation

$$J[\cdots, n_{e,i} = 2, \cdots] = -J[\cdots, n_{e,i} = 0, \cdots] - (2\nu_e + 1)J[\cdots, n_{e,i} = 1, a_r \rightarrow a_r - 1, b_e \rightarrow b_e + 1, \cdots], \quad (2.3)$$

so every massive endpoint state reduces to the representative set $\{0, 1\}$. The package applies this reduction by construction: the generated seeds contain only $n_{e,i} \in \{0, 1\}$.

Massless quotient. A massless full line does not distinguish its two endpoints. The package fixes an ordered state $(n_{e,1}, n_{e,2})$ per line and works with the quotient

$$F_{01} = -F_{10}, \quad F_{11} = -F_{00}, \quad (2.4)$$

so the public family keeps a single bit per massless full line, taken from the representatives F_{00} and F_{10} . This is an exact algebraic identification, not a sampling prescription.

Shrink rules. A massive line may be shrunk only in the `massiveFull` type and only when $n_{e,1} + n_{e,2} = 1$. The shrink multiplies the sector by the coefficient $-\chi_e(-1)^{n_{e,r}}C_e$, where $\chi_e = \pm 1$ is fixed by the two vertex types and $C_e = \frac{4i}{\pi}e^{\pi \text{Im} \nu_e}$. A massless full line shrinks with a ∓ 2 delta term and a $+2$ term, without a Wronskian prefactor and without label shifts. If all lines meeting at a vertex shrink simultaneously, the odd-subset contact terms

$$\sum_{\substack{\emptyset \neq S \subseteq B \\ |S| \text{ odd}}} 2^{1-|S|} \prod_{e \in S} D_e \prod_{e \in S} J_e \quad (2.5)$$

are generated. Shrunk lines never reappear as denominators. Their magnitude powers live in the structured sector prefactor.

2.3 Kinematics

The structural loop number L is the first Betti number of the internal undirected multigraph. Let E count the internal edges: parallel edges are counted once per line, and each

self-loop contributes one edge. External legs do not contribute to E . Let V count the vertices and C the connected components. Then

$$L = E - V + C. \quad (2.6)$$

Together with K independent loop-related external momenta, the loop number fixes the number of scalar products,

$$N_{\text{sp}} = \frac{L(L+1)}{2} + LK. \quad (2.7)$$

The user declares two separate lists. The first list, `loopExternalMomenta` $\{k_1, \dots, k_K\}$, defines the loop coordinates $\mathbf{ss}ij = \sqrt{\text{sp}[k_i, k_j]}$ of the loop Gram matrix. The second list, `independentExternalMomenta`, defines the no-loop magnitudes $|\mathbf{kE}i|$ that appear in vertex energies and fixed-line masses. These magnitudes are parameterized by $\mathbf{sE}i$. Only magnitudes are coordinates. The package never outputs $\text{sp}[k_{Ei}, k_{Ej}]$, and it never eliminates intermediate magnitudes by a cosine rule. Magnitudes that depend on the declared ones are stored as explicit bindings. They still contribute to vertex phases through the chain rule.

User-defined ISPs are declared as `name/expr/range` associations, for example

```
<|"name" -> rho1, "expr" -> sp[q1, k], "range" -> {0, 1}|>,
```

and they must form an invertible coordinate system on the space of scalar products together with the propagator squares. The definition zero point of an ISP is fixed at 0: positive powers are numerator powers, and a negative power means an extra denominator. Kinematic rules of the form $\text{sp}[k_i, k_j] \rightarrow \text{dynamical variables}$ are validated by the two-stage interface `DSKinematics/DSInit`. The interface reports exact, over-complete or under-complete states. Only the exact state opens derivative generation.

2.4 Parity structure

The characteristic discrete invariant of de Sitter IBP is the parity of the sum of the line labels. For every cyclic line e the quantity

$$p_e = b_e + n_{e,1} + n_{e,2} \pmod{2} \quad (2.8)$$

is invariant under all loop and time generators acting on a full massive line. It is preserved by the h/H shrink of massive lines. For massless full lines it flips by construction: the contact operation maps $b_e + n_{e,1} + n_{e,2} \rightarrow b_e + 1$, so the massless parity of a shrunk sector is offset by one. Cross lines carry no parity constraint. Every generator preserves the parity vector $(p_e)_e$. The IBP system of a family therefore splits into independent parity blocks. For families with user-provided `parityConstraints` the package restricts the seeds to the requested block, for example $b_{[1]} + n_{[1,1]} + n_{[1,2]} \equiv 0$. This reproduces the parity-split IBP system of ref. [8]. ISP indices receive no parity filter.

2.5 Dynamical rules and massive function systems

Every massive line is assigned a second-order function system $f_e'' + P_e(x) f_e' + Q_e(x) f_e = 0$. The default "h" preset uses $P_h = (2\nu_e + 1)/x$ and $Q_h = 1$. The package also accepts bare Hankel functions H and general user systems (P, Q, T, W) . Let T_e be the transport matrix of the fundamental system and W_e its Wronskian. The package tracks the combination $W_T = \det(T) W$ through all shrinks. The theta-coincidence and Wronskian shrink terms used by the time-IBP generators are derived from it. Ordinary derivatives of dynamical quantities never generate theta shrinks and never invoke W_T . The user chooses the convention once at initialization. The IBP generators, the derivative operators and the tree-level track all read the same compiled function system.

2.6 Tree-level and time-only families

For graphs without cycles the package exposes a `timeOnly` family with the unique public representation

```
J[sectorKey_String, timeShifts_List, stateBits_List]
```

Here `sectorKey` records, per root line, whether the line is full (1) or shrunk (0). The list `timeShifts` gives the time powers a_v of the active vertices. The list `stateBits` gives one bit per massive endpoint or massless quotient state, in the frozen root-line order. There are no momentum generators, no propagator powers and no ISPs. Fixed-line magnitude powers remain in the structured sector prefactor and still enter derivative generation. The massive-only time-only families are fully certified. Families that contain massless full lines are returned as `PendingRederivation`. The reason recorded with them is `masslessQuotientFormulaNotCertified`. The public interface does not bypass this gate.

3 Using the package

3.1 Initialization

A case is a named topology declaration. Initialization runs a structural audit of the loop number, the vertex and line registry and the external-momentum routing. It then runs the kinematic audit of section 2.3. The result is a context that every later command consumes:

```
context = DSInit[case, KinematicRules -> custom];
```

The optional two-stage interface `DSKinematics[case]` first returns a proposal with the default rules and a selection template. The user can then inspect the kinematic coordinates `ssij`, `sEi`, the momentum-routing audit and the capability flags before committing. Under-complete declarations abort the initialization. Over-complete ones allow symbolic seed generation but disable derivatives and the inner-form rewriting. In the over-complete case common loop shifts are quotiented out through an affine basis `effectiveLoopExternalMomenta`.

3.2 Seeds and the IBP envelope

`DSSeeds[context]` returns the canonical seed templates of the family. Every actual sector receives all active time generators and all $L(L + K)$ loop-momentum generators. The massive endpoint states are pre-reduced by the equation of motion (2.3). Massless full lines run over the two quotient representatives. The discrete n labels are eliminated, and the analytic seeds keep general continuous labels $a_v, b_e, b_{S,e}$ and general ISP indices. Each template is hash-sealed. Templates that the equation of motion and the canonicalization reduce to the identity $0 = 0$ are retained and audited rather than dropped.

The continuous labels are swept by `DSGenerateIBP`. The unified form

```
ibpData = DSGenerateIBP[allSeeds, {-1, 1}];
```

requires that every root continuous index lies in the closed integer interval $[-1, 1]$ *after* all shifts of the generated relations. The package inverts the per-index shift sets and derives the narrower seed ranges automatically. The per-index form

```
ibpData = DSGenerateIBP[allSeeds,
  {a[v1], -1, 1}, {a[v2], 0, 1},
  {b[e1], 0, 2}, {b[e2], -1, 1}];
```

must cover all root continuous indices exactly. ISP lower bounds are combined with the user target by taking the larger lower bound, so the package never extends a range on its own in order to cover numerator terms. The output is a sealed contract with status `"generated"`, a template-set hash, an integrity audit and the capability flags `sealedQ`, `completeSystemQ` and `subsetQ`. Only sealed complete systems may proceed to a formal reduction.

3.3 Linear systems and external reduction

`DSLinear[ibpData, context]` converts the sealed seeds into a backend-neutral association `linearData`. Its fields are the equation list `linearEquations`, the master-integral ordering `integrallist`, the pivot rules `integralRules`, the sector metadata `sectorMetadataList` and the readiness reports. The order of `integrallist` fixes the master-integral ordering. The command `DSReorderIntegrals` re-indexes it if the user prefers a different order. The two-stage Kira plan is built purely in memory:

```
prePlan   = DSKiraPlan[linearData, <|"stage" -> "preReduction", ...|>];
preExport = DSKiraExport[prePlan];
(* ... external Kira run identifies the master integrals ... *)
userMIData = DSUserMI[linearData, userExpressions, <|
  "names" -> activeNames, "activeIndices" -> activeIndices,
  "derivativeVariables" -> {ss11, P0}, "scalingDegrees" -> degrees|>];
formalPlan = DSKiraPlan[linearWithUserMI, <|"stage" -> "formal", ...|>];
formalExport = DSKiraExport[formalPlan];
(* ... external Kira run reduces to the user basis ... *)
importedData = DSKiraImport[formalWorkspace];
```

The pre-reduction stage freezes the candidate targets and lets the external reducer discover the masters. The command `DSUserMI` then attaches the user basis. It checks the global and the active rank, and it stores reversible pivot rules. The formal stage constructs the analytic derivative closure of the active basis *before* any export. It refuses to write if internal helper symbols survive. Export manifests carry the SHA-256 identity of every artifact, and `DSKiraImport` re-verifies the contents instead of trusting version strings. The serializer maps momenta to the imaginary axis $k \rightarrow -I\,ik$ expected by Kira. The same `linearData` feeds the Rational Tracer [11] pipeline through the equation-system interface.

3.4 Differential equations and checks

After the reduction, `DSDE` differentiates the master integrals with respect to the independent variables. Loop-Gram coordinates are handled through external-vector derivative decompositions. No-loop magnitudes are handled through line-local radial derivatives, including their dependent bindings. Vertex energies enter as scalar parameters. Explicit coefficients are differentiated directly, while bare integrals are shifted and canonicalized. The result is reduced back to the active basis. For the pure massive bubble the resulting equations are in d log form. At the recorded exact point they reproduce the matrices of ref. [8] entry by entry.

Every integral of the family is a homogeneous function of the dimensionful variables. Let $x_1, \dots, x_m$ denote these variables, that is, the loop coordinates `ssij`, the no-loop magnitudes `sEi` and the mass scales. Each master integral I_k then satisfies the Euler relation

$$\sum_{a=1}^m x_a \frac{\partial I_k}{\partial x_a} = d_k I_k, \quad (3.1)$$

where d_k is the constant scaling degree of I_k . In matrix form, with $\partial I / \partial x_a = A_a(x) I$, the identity (3.1) becomes

$$\sum_{a=1}^m x_a A_a(x) = \Delta, \quad \Delta = \text{diag}(d_1, \dots, d_N). \quad (3.2)$$

The command `DSScaleCheck` verifies (3.2) symbolically on the reduced basis. It also checks the source-level time and loop-momentum IBP identities before the reduction. Because Δ is constant and diagonal, (3.2) is a strong self-consistency test of the whole chain from seeds to differential equations. The tree-level track exposes `DSTreeDLogDE` for direct d log differential equations.

At the atomic level the package also exposes the generators themselves:

```
dtau[i, expr, context]      (* time generator at vertex i *)
dqq[i, j, expr, context]    (* d/dq_i . q_j *)
dqk[i, j, expr, context]    (* d/dq_i . k_j *)
ds[expr, var, context]      (* full variable derivative *)
```

together with the form rewriters `rep2innerform`, `rep2outform` and `rep2Integrand`. The last command expands a J into its explicit integrand without integrating. The short signatures rely on the single registered topology. Explicit contexts should be used when several topologies are alive.

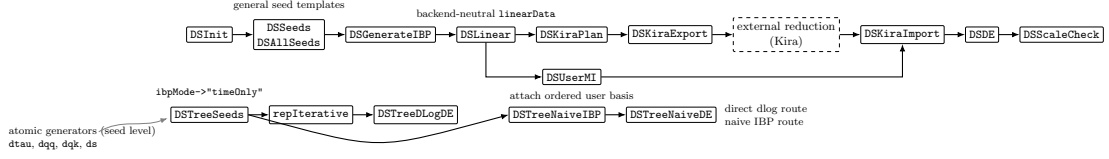


Figure 3. The public workflow of dSIBP. Full (loop) families run from DSInit through the seed, envelope and linear-system stages to the two-stage Kira export and import and to the differential-equation layer. The command DSUserMI attaches the user basis between the two reduction stages. Tree-level (timeOnly) families run through DSTreeSeeds and repIterative to DSTreeDLogDE, with the naive IBP route as an independent cross-check. The atomic generators dtau, dqq, dqk and ds are available at every stage.

3.5 Tree-level track

The tree-level functions mirror the loop workflow:

```
treeSeeds = DSTreeSeeds[context];
iterative = repIterative[treeSeeds, context];
naiveIBP = DSTreeNaiveIBP[treeSeeds, context];
naiveDE = DSTreeNaiveDE[naiveIBP, context];
dlogDE = DSTreeDLogDE[iterative, context];
```

The iterative route substitutes the general derived formulas directly. The naive route reduces raw sampled IBP relations. The two routes must agree, and the package reports the flag `deRoutesAgree`. The tree-level differential equations have also been compared with the package MadStree [12]. MadStree evaluates tree-level cosmological correlators and builds its own d log differential equations. For a mixed three-vertex tree family with fifteen masters, the parameter derivatives of dSIBP were reduced by MadStree for the five variables $k_1, k_2, k_3, \mathbf{sE1}, \mathbf{sE2}$. All seventy-five derivatives closed, and the five 15×15 matrices agree entry by entry with the direct d log formula route of MadStree. Figure 3 summarizes the two tracks and their connection to the external reducers. Appendix A lists the public functions with one-line descriptions.

4 Examples

The package ships six worked examples. Figure 4 shows the four graphs discussed below. The quoted Kira times are wall-clock times of the external reduction on a single workstation. The Kira inputs were written by DSKiraExport.

Sunrise (example 03). The graph of figure 4a has two vertices and three parallel lines, hence $L = 2$. Line 1 is a massive full line of mass ν_1 , and lines 2 and 3 are massless full lines. The loop momenta are l_3 and k_{321} , and the single loop external momentum is k_L . Thus $K = 1$, and the loop Gram matrix gives the coordinate $\mathbf{ss11} = |k_L|$. No no-loop magnitude is declared: both vertices carry the same external-leg energy k_E , which is the second derivative variable. The top sector receives two time generators and $L(L + K) = 6$

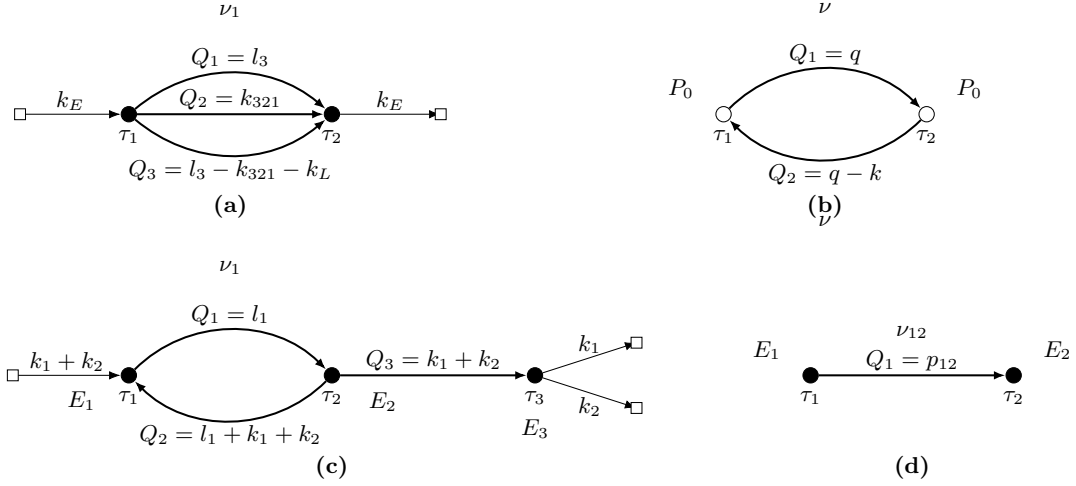


Figure 4. Example graphs. The fill colour of a vertex encodes its branch sign: black is $+$ and white is $-$. (a) The two-loop sunrise of example 03: one massive full line of mass ν_1 and two massless full lines between two $+$ vertices; both vertices carry the same external-leg energy k_E . (b) The pure massive bubble of example 04: two massive full lines of the same mass ν between two $-$ vertices with vertex energy P_0 ; the loop external momentum is k . (c) The mixed bubble-plus-tree case of example 06: a massive full line of mass ν_1 and a massless full line form the loop between v_1 and v_2 , and a massless bridge connects v_2 and v_3 ; the external legs carry $k_1 + k_2$, k_1 and k_2 . (d) The two-vertex tree of example 05: a massive line of mass ν_{12} whose momentum p_{12} is a no-loop magnitude.

loop-momentum generators. Two ISPs are declared, $\text{sp}[k_{321}, l_3]$ and $\text{sp}[l_3 - k_{321} - k_L, l_3]$, each with range $\{0, 1\}$. The symmetry that swaps the two massless lines exchanges them, and the package canonicalizes the two ISPs together with that swap. The example stops after the general stage. It generates the canonical seeds and the general parameter-derivative operators, and it neither samples the continuous labels nor runs a reduction, a differential equation or a scaling check. The seeds are checked by applying **ds** to the top-sector integral with respect to both variables.

Pure massive bubble (example 04). The bubble of figure 4b has two vertices on the $-$ branch. Its two massive full lines carry the same mass ν and the loop momenta q and $q - k$. The loop external momentum k gives the loop coordinate $\text{ss11} = |k|$. Both vertex energies are set to the parameter P_0 , which is the second derivative variable. This example runs the full closed loop: seeds, envelope sweep, linear system, two-stage Kira export, external reduction, import, differential equations and scaling check. The Kira workspace must lie outside the repository, and the package itself never starts Kira. The linear system is written at a fixed rational probe point for the dimension and the masses, while the differential equations keep ss11 and P_0 symbolic. The reduction closes with no unreduced target and returns nineteen master integrals. The two 19×19 differential-equation matrices in $d \log$ form reproduce those of ref. [8]: at the recorded exact point each of the three variable conventions agrees entry by entry, 361/361. The scaling identities (3.2) hold symbolically on the same basis, with 33/33 source-level identities and 19/19 active-basis checks.

Mixed bubble plus tree (example 06). The most demanding released case is figure 4c. Lines 1 and 2 form the loop: line 1 is a massive full line of mass ν_1 , and line 2 is a massless full line. Line 3 is a massless bridge between v_2 and v_3 . It lies outside the loop, so its pack carries no denominator power. The external legs carry $k_1 + k_2$ into v_1 and k_1, k_2 into v_3 . The loop external momentum is $k_1 + k_2$, which gives `ss11`. The independent no-loop magnitudes k_1, k_2 give `sE1, sE2`. Both cycle lines are restricted to even parity, $b_{[e]} + n_{[e,1]} + n_{[e,2]} \equiv 0$ for $e = 1, 2$, while the fixed bridge is excluded from the parity constraints. The parity-restricted family has eighty-one master integrals, and the generated system contains 818,217 relations. The external Kira reduction takes about 160s and closes with no unreduced target. The package then produces six 81×81 differential-equation matrices, one for each of the variables `ss11, sE1, sE2, E1, E2, E3`, with no residual symbols. All 81/81 source-level Euler and IBP identities vanish exactly. The reduction-based scaling certificate is evaluated at the recorded exact point, because all six scaling variables are fixed in the formal plan. The example also exercises the parameter redefinition `DSRedefineParameters`, a bound-energy coordinate binding $E_0 = |k_1| + |k_2|$, and the over-complete and under-complete kinematic audits.

Two-vertex tree (example 05). A massive line of mass ν_{12} connects two $+$ vertices, figure 4d. There is no loop momentum. The momentum p_{12} of the line is declared as an independent no-loop magnitude, so the package builds the coordinate `sE1 = |p12|`. The `timeOnly` family has the two sector keys "1" and "0", for the full and the shrunk line. The iterative route and the naive IBP route give the same $d \log$ differential equations in $E_1, E_2, sE1$, so `deRoutesAgree` is `True`. The same example also declares a one-line massless family. Its quotient states reduce to $n_{1,1} \rightarrow 0$, $n_{1,1} \rightarrow 1$ and $n_{1,2} \rightarrow 0$ over the sector keys "0" and "1". Its formula route returns `PendingRederivation` with the reason `masslessQuotientFormulaNotCertified`, as required by the gate of section 2.6.

Mixed bubble workflow and bare Hankel system (examples 01, 02). Example 01 declares a bubble with one massive and one massless full line between two $+$ vertices. It runs the unified public entry through the discrete templates, the continuous sweep and the linear system. It builds the pre-reduction plan in memory without exporting or running Kira. Example 02 supplies a bare Hankel function system (P, Q, T, W) instead of the "h" preset. The package compiles it through the same public workflow and generates the seeds of the corresponding family. Together the two examples cover the initialization, the envelope sweep and the function-system interface of sections 3 and 2.5.

5 Summary and outlook

dSIBP turns the canonical IBP structure of de Sitter time integrals into a practical MATHEMATICA tool. From a graph declaration it produces sealed canonical seeds, backend-neutral linear systems, Kira and Rational-Tracer export manifests with artifact identities, differential equations and scaling checks. It covers arbitrary topologies that mix massive and massless propagators of all four Schwinger–Keldysh types. The parity-split structure, the massive equation-of-motion reduction and the massless quotient are applied by construction, so the user never handles raw label spaces. Ongoing work concerns the certification of the massless

full-line tree-level formulas, which are currently gated behind `PendingRederivation`. Further directions are a deeper integration with hypergeometric solution methods [7] and the connection to numerical evaluation pipelines such as MadStree [12] and to high-precision series-transport solvers [16].

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We thank the developers of Kira [9, 10], Rational Tracer [11] and MadStree [12] for making their tools available.

A Public functions

Function	Description
DSInit	initialize a case; structural and kinematic audits; returns the context.
DSKinematics	two-stage kinematic proposal and validation.
DSPParameterNotation	inspect the current parameter notation.
DSRedefineParameters	replace the dynamical rules; re-audits and rehashes.
DSSeeds	canonical seed templates of a family.
DSAllSeeds	flattened seed template list.
DSSeedGroups, DSSeedGroupMetadata	grouping of the templates by sector, class and generator.
DSMetaSeedRange	per-group shift metadata for envelope planning.
DSGenerateIBP	sweep continuous indices; sealed IBP relations.
DSLlinear	backend-neutral linear system (<code>linearData</code>).
DSReorderIntegrals	re-index the master-integral ordering.
DSKiraPlan	in-memory pre-reduction or formal reduction plan.
DSKiraExport	write Kira input directories; manifest with SHA-256 identities.
DSKiraImport	read back external reduction results with identity checks.
DSUserMI	attach a user master-integral basis with rank checks.
DSDE	differential equations of the reduced basis.
DSScaleCheck	Euler scaling identities of the family.
DSTreeSeeds	<code>timeOnly</code> seed templates.
repIterative	tree route via the general derived formulas.
DSTreeNaiveIBP, DSTreeNaiveDE	tree route via raw sampled IBP.
DSTreeDLogDE	tree $d \log$ differential equations.
dtau, dqq, dqk, ds	atomic generators and the full variable derivative.
rep2innerform, rep2outform	notation rewriters between user and internal forms.
rep2Integrand	expand J into the explicit integrand.
symmetry	apply user and gated self-loop symmetry rules.

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